



Module Code: ITS61504 (Feb 2026)

Module Name: Data Mining

Assignment No./Title	Assignment (Individual) 30% (ASSIGNMENT 2 – EXECUTIVE SUMMARY – MLO2)
Course Tutor/Lecturer	Ts. Dr Anbuselvan Sangodiah
Section No	[Provide your Section No]
Student Name, ID and Signature	
Student Name	
Student ID	
Signature	

Declaration (need to be signed by the student; otherwise, the assessment will not be evaluated)

Certify that this assignment is entirely my own work, except where I have given fully documented references to the work of others, and that the material contained in this assignment has not previously been submitted for assessment in any other formal course of study. I have also read and accept the content of the **documentation and submission instructions section** of the assignment 2 question.

Marks/Grade:	Evaluated by:
Evaluator's Comments:	
* Please include this cover page for your project submission	

EXECUTIVE SUMMARY

PURPOSE (MLO2)

The purpose of this Assignment 2 is to:

1. **MLO2:** *Communicate findings and insights derived from data mining processes effectively to both technical and non-technical stakeholders for a given scenario.*

Objectives

- **Implementation of Apriori Algorithm: (Must use the dataset provided in Assignment 1)**
 - o Use the Apriori Algorithm using *Python* to apply the association analysis technique. Document the steps taken in implementing this algorithm (*screenshot Python codes when documenting the steps*). You may use the conventional method (*developing Python scripts by a developer*) and the LLM framework with advanced libraries to perform data preprocessing/evaluation only.
 - o Document frequent item sets and association rules
- **Implementation of FP Algorithm: (Must use the dataset provided in Assignment 1)**
 - o Use the FP Algorithm using *Python* to apply the association analysis technique. Document the steps taken in implementing this algorithm (*screenshot Python codes when documenting the steps*). You may use the conventional method (*developing Python scripts by a developer*) and the LLM framework with advanced libraries to perform data preprocessing/evaluation only.
 - o Document frequent item sets and association rules
- **Summary of Observations:**
 - o Provide detail explanation from the findings to the recommendations/conclusions.
 - o Compare both algorithms when reporting interesting patterns from the findings.
- **Report Writing:**
 - o Present your findings in a structured and professional manner, ensuring clarity for both technical and non-technical audiences. Also, adherence to proper file naming. This includes the proper submission of Python code.
 - o **Deadline:** Ensure that the assignment is submitted by the due date via *MyTimes*.

You are required to carry out this Assignment 2 assignment individually. Generally, in this assignment, you will be analyzing the same data set provided in Assignment 1 by implementing Apriori and FP algorithms to carry out association analysis, inferring interesting patterns from the findings, and making good recommendations to achieve the set objectives.

EXECUTIVE SUMMARY (30%)

Report Outline

1. Implementation of Apriori Algorithm
2. Implementation of FP Algorithm
3. Summary of Observations

Deliverables

1. A well-structured and properly formatted academic document that contains the detailed specifications for Assignment 2.
2. Python codes (PyCharm Project File/Python Files)

***Deliverable 1** must be submitted in two formats: *doc* and *pdf*

File format: **AssignmentPart2_Report_SectionNo_IDNo**

***Deliverable 2** must be submitted in a zip format – file format

File format: **AssignmentPart2_Codes_SectionNo_IDNo**

Assignment 2 Due Date: 6/3/2026 (Friday by 12.00 am - MYT) submit via mytimes.taylors.edu.my submission link.

Important Milestones for your assignment work (Assignment 2)

% of completion work of assignment (CW)

Milestone 4 (Week 4) – CW - 50%: Completion of implementing association analysis algorithms

- Preparation of data set for association analysis.
- Application of association analysis algorithms.

Milestone 5 (Week 5) – CW - 100%: Completion of inferring interesting patterns.

- Identification of frequent items sets and association rules.
- Completion of inferring interesting patterns from association analysis.
- Preparation for Assignment 2 report submission.

Documentation & Submission Instructions

As you prepare your assignment for submission, please adhere to the following guidelines to ensure your work is clear, well-documented, and demonstrates your coding process effectively:

1. Use the provided front page template for your assignment. It should include the full details of yourself. Make sure all information is accurate and clearly presented.
2. As for documenting Python code, accompany every segment of code with detailed comments. These should explain the function and mechanism of the code, the expected outputs, and any important considerations or potential issues that might arise.
3. The academic Honors Code applies to this assignment. Plagiarized work would not be marked.
4. All assignments are due on the dates given by the instructor on *MyTimes* or on a certain day of the week as specified in the lesson plan.
5. All work must be original, and if taken from any works other than yours, must be properly referenced using the APA Referencing System.
6. Before proceeding to the final submission, you must first submit your PDF document to Turnitin for a plagiarism check. Ensure you review the similarity report (*similarity rate should be below 25%*) and make any necessary adjustments to your work to adhere to academic integrity guidelines. **Attach the one-page Turnitin report to the assignment report.**
7. Submit the **Assessment Submission Declaration Form** along with the assignment report. **(Zip this form along with the assignment report)**
8. You **MUST** submit your assignment via MyTimes only.
9. All information, figures, and diagrams obtained from external sources must be referenced using the APA referencing system accordingly.
10. Plagiarism and collusion are serious offences, and all students involved will automatically be awarded zero marks.
11. Take note: Submission of the assignment **DOES NOT** mean that you pass the assignment by default. Reasons for failing to submit assignment in a timely manner due to computer crash/repair, not feeling well are not valid.
12. Final Review: Before submission, review your PDF/Word document to confirm that all parts of the assignment are included and that the document is not missing any content.

13. Incomplete submission:

- a. Examples of incomplete submission – failing to submit source codes, reports, *Turnitin* report, details are incomplete (cover page), etc.
- b. The penalty for failing to submit either source code or report or both, is zero marks, and for other types of incomplete submissions, the penalty is that your assignment marks will be capped at **B- (grade)**.

14. Failure in Compliance:

- a. Examples: failing to observe similarity rate, APA referencing system, etc.
- b. The penalty is that your assignment marks will be capped at **B- (grade)**.

15. Late submission:

- a. Late submission (*1 to 3 days – including non-working days*) – assignment marks will be capped at **B- (low end - 60)**
- b. Late submission (*4 to 6 days – including non-working days*) - assignment marks will be capped at **C (low end - 50)**
- c. Any late submission on the 7th day or after (*including non-working days*) – assignment marks will be zero

MARKING SCHEME – ASSIGNMENT 2

Items	Marks
Implementation of Apriori Algorithm	8
Implementation of the FP Algorithm	8
Summary of Observations	10
Overall documentation	4
TOTAL	<u>30</u>

MARKING RUBRICS – ASSIGNMENT 2

Criteria	8-10 marks	6-7 marks	5 marks	1-4 marks	0 marks
Implementation of Apriori Algorithm	Correct and efficient implementation of the Apriori algorithm. Handles large datasets effectively. Includes optimizations (e.g., candidate generation, pruning). Well-documented and commented code. Application of conventional and LLM methods in data preprocessing/evaluation is evident.	Correct implementation of the Apriori algorithm. Good at handling large datasets. Includes some basic optimizations. Code is reasonably well-documented.	Basic implementation of the Apriori algorithm. May have minor errors or inefficiencies. Limited or no optimizations. The code has minimal documentation.	Incorrect or incomplete implementation of the Apriori algorithm. Does not handle datasets effectively. No optimizations implemented. Code is poorly documented or missing.	No attempt to implement the Apriori algorithm.
Implementation of the FP-Growth Algorithm	Correct and efficient implementation of the FP-growth algorithm. Handles large datasets effectively. Efficient construction of the FP-tree. Well-documented and commented code. Application of conventional and LLM methods in data preprocessing/evaluation is evident.	Correct implementation of the FP-growth algorithm. Good at handling large datasets. Construction of the FP-tree is reasonably efficient. Code is reasonably well-documented.	Basic implementation of the FP-growth algorithm. May have minor errors or inefficiencies. Construction of the FP-tree may be inefficient. The code has minimal documentation.	Incorrect or incomplete implementation of the FP-growth algorithm. Does not handle datasets effectively. Construction of the FP-tree is inefficient or incorrect. Code is poorly documented or missing.	No attempt to implement the FP-growth algorithm.
Summary of Observations	Clear and concise summary of the findings. Analysis of the performance of both algorithms in terms of frequent item sets, association rules, (e.g., runtime, memory usage). Recommendations are significant. Comparison of the results obtained from both algorithms. Discussion of the strengths and weaknesses of each algorithm.	Adequate summary of the findings. Basic analysis of the performance of both algorithms. Some comparison of the results. Recommendations are decent. Limited discussion of strengths and weaknesses.	Basic summary of the findings. Limited analysis of algorithm performance. Limited comparison of results. Limited discussion of strengths and weaknesses.	Incomplete or inaccurate summary of the findings. No analysis of algorithm performance. No comparison of results. No discussion of strengths and weaknesses.	No summary of observations provided.
Overall documentation	Well-organized and professional presentation of the project. Clear and concise writing style. Properly formatted and cited sources. Effective	Good organization and presentation of the project. Clear and generally concise writing. Properly formatted sources. Effective	Adequate organization and presentation of the project. Some clarity and conciseness in writing. Some	Poor organization and presentation of the project. Unclear and difficult to read. Significant formatting and citation issues.	Poorly organized and presented project. Unclear and difficult to understand. No proper formatting or citations. No

	use of tables, figures, and diagrams.	use of some tables, figures, and diagrams.	formatting and citation issues. Limited use of tables, figures, and diagrams.	Ineffective use of tables, figures, and diagrams.	use of tables, figures, or diagrams.
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